

# 完整案例透视 GraphRAG 工作机制：从图构建到图检索



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## 微软 GraphRAG 工作原理解析：从图构建到搜索

### 引言

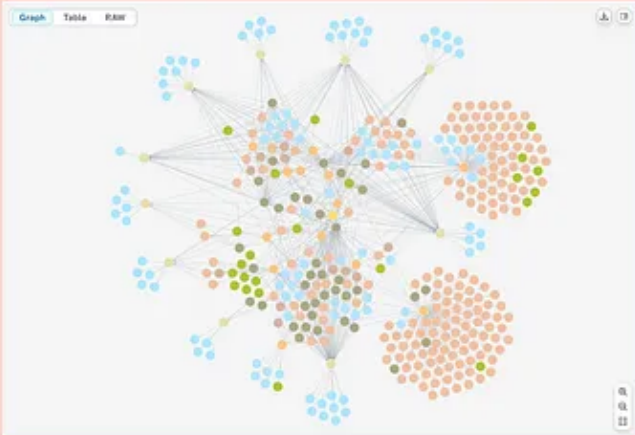
本文基于微软研究论文《*From Local to Global: A GraphRAG Approach to Query-Focused Summarization*》及其开源项目，深入剖析了 GraphRAG 的工作原理。传统的检索增强生成（RAG）在处理分散、复杂且互相关联的信息时常常力不从心。GraphRAG 通过引入知识图谱，旨在解决这一核心问题，它能够从非结构化数据中构建实体和关系网络，从而提供更具深度和上下文感知的答案。本文将通过一个真实案例，逐一拆解 GraphRAG 从图构建到图查询的每一个关键步骤，帮助你彻底理解其内部机制。

### 什么是 GraphRAG?

简而言之，GraphRAG 是一种利用图结构来增强检索增强生成（RAG）的技术。本文将重点关注微软的实现方案，其核心流程可分为两大阶段：

- 图构建（Graph Creation / Indexing）：将原始文档转化为知识图谱。
- 查询（Querying）：利用构建好的图谱来回答用户问题，主要包括局部搜索（Local Search）和全局搜索（Global Search）两种模式。

为了具体说明，我们将以西班牙小说《Penitencia》为例，演示 GraphRAG 如何对其进行索引和查询。

| Step 1: Create the Graph                                                            | Step 2: Query the Graph                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  | <b>Local Search</b><br><br>Q: "What kind of retribution is Laura seeking, and why?" |
|                                                                                     | <b>Global Search</b><br><br>Q: "What are the themes of the book?"                   |

### 第一阶段：图构建（索引）

图构建过程始于运行 `graphrag index` 命令，它触发了两个核心模块：实体提取和图分区。这一阶段的目标是将非结构化的文本数据转换成一个结构化的、包含丰富上下文的知识图谱。



## 图构建 (Graph Creation)

源文档 (如 《Penitencia》)

1. 文本分块

文本单元 (Text Units)

2. LLM提取

实体与关系

3. 去重与总结

最终实体/关系描述

4. 构建图

NetworkX 图对象

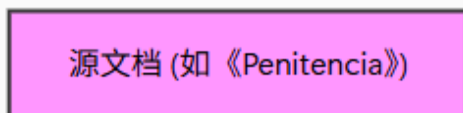
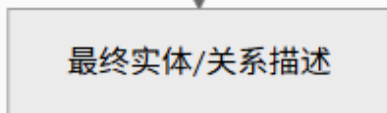
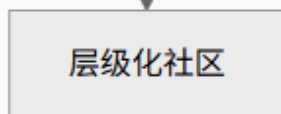
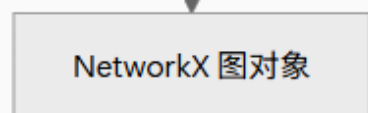
5. Leiden算法分区

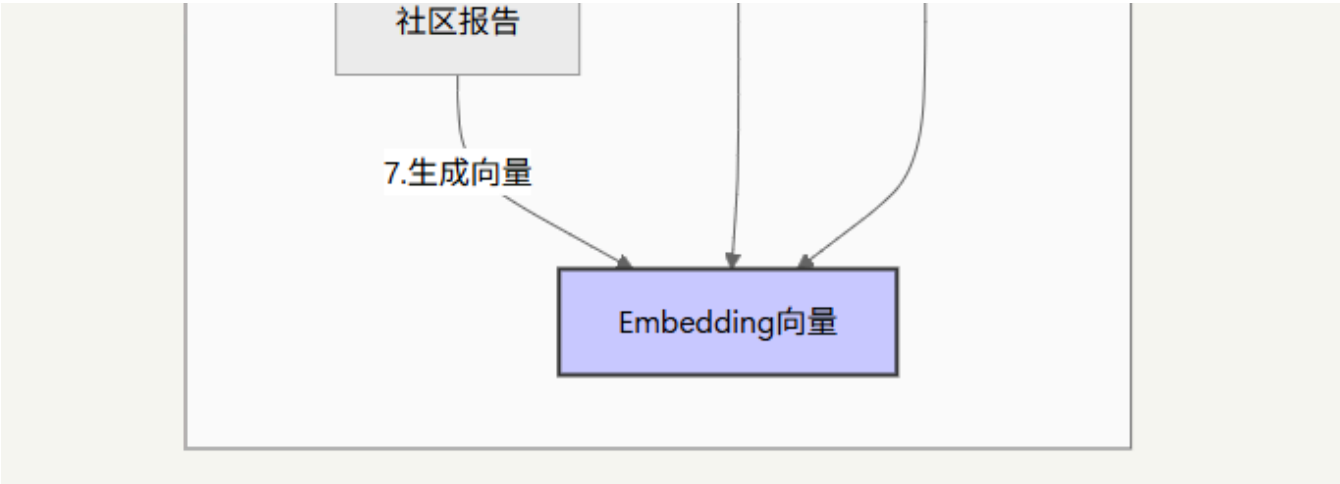
层级化社区

6. LLM生成报告

7. 生成向量

7. 生成向量





实体提取

实体提取是将文本转化为图谱节点和边的基础。

(1) 文本分块 ( create\_base\_text\_units )：首先，系统将原始文档（如《Penitencia》）分割成固定大小的文本块（Text Chunks），每个块拥有唯一的ID。

| id                              | text                                  | document_ids                                                          | # n_tokens |
|---------------------------------|---------------------------------------|-----------------------------------------------------------------------|------------|
| b65bb9f86e8bf83b6a5c7e814c2bfb7 | Cap&#237;tulo 1 1 Llevaba recorrid    | ['6852a66a08b38d4d412f043ca78ad                                       | 1200       |
| dc1d18ef85ca582714aa0a57b840c7  | su mensaje tal y como lo habían dise  | ['6852a66a08b38d4d412f043ca78ad                                       | 1200       |
| 87a014234374c8bc75a97357cac08e  | una de hierro industrial imitando ant | ['6852a66a08b38d4d412f043ca78ada50c66b2                               | 1200       |
| da5b98d5d131e88a13bc6e542b796   | . En uno de los frentes de pared, per | ['6852a66a08b38d4d412f043ca78adbb5e4724a4a94fb2849d62ad29f0000613f83d | 1200       |
| 0635907245d0f984d40739d87dd47c  | con la habitación cochambrosa que     | ['6852a66a08b38d4d412f043ca78ad64c64fd090dda6']                       | 1200       |

(2) 实体与关系提取 ( extract\_graph )：接着，利用大语言模型（LLM）分析每一个文本块，根据预设的提示词（Prompt）提取出其中的实体（Entities）和关系（Relationships）。默认情况下，实体类型包括地理位置（Geo）、人物（Person）、事件（Event）和组织（Organization）。

在这个阶段，可能会出现重复的实体和关系。例如，主角“Jon”在 82 个不同的文本块中被提及，因此他会被提取 82 次。

| title                        | type         | description                              | text_unit_ids                                                                                                                                                                                                                                                               | # frequency |
|------------------------------|--------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| MADRID                       | GEO          | ['Madrid is the capital city of Spain, f | ['b65bb9f86e8bf83b6a5c7e814c2bfb                                                                                                                                                                                                                                            | 6           |
| JON                          | PERSON       | ['Jon is the protagonist who is drivin   | ['b65bb9f86e8bf83b6a5c7e814c2bfb                                                                                                                                                                                                                                            | 82          |
| LA TRAVIATA                  | EVENT        | ['La Traviata is a                       | ['6852a66a08b38d4d412f043ca78ad                                                                                                                                                                                                                                             | 1           |
| COCHE DEPORTIVO              | ORGANIZATION | ['The sports car                         | ['Jon is the protagonist who is driving away from Madrid, reflecting on his life choices and seeking change.', 'Jon is a character who is contemplating a media deal and has a close friendship with Julio, his representative.', 'Jon is a character who explores a modern | 1           |
| REDES SOCIALES               | ORGANIZATION | ['Social media p                         | ['Jon is a character who is seeking to create a                                                                                                                                                                                                                             | 1           |
| CONTRATO                     | EVENT        | ['The contract i                         | luxurious living space while maintaining a facade of a dilapidated exterior to avoid suspicion.', 'Jon is a character who explores a modern and luxurious home, reflecting on his experiences and changes in life.',                                                        | 1           |
| AGENCIA DE PUBLICIDAD        |              | ['']                                     | ['Jon is an individual who                                                                                                                                                                                                                                                  | 1           |
| JULIO                        | PERSON       | ['Julio is Jon's b                       | ['Jon is a character reflecting on his life and experiences, particularly in relation to his living space and past interactions with the world.', 'Jon is a character who is having a conversation with his mother over the                                                 | 21          |
| CUATRO LATAS                 | GEO          | ['Cuatro Latas i                         | phone, dealing with personal changes and challenges.', 'Jon is a character who is having a conversation with his mother about his new                                                                                                                                       | 1           |
| FÉLIX RODRÍGUEZ DE LA FUENTE | PERSON       | ['Félix Rodrígue                         | living situation and the house he has bought.', 'Jon is an individual who is living in secrecy and has chosen to isolate himself from others, communicating only with his family and a co                                                                                   | 1           |
| MERCADO                      | GEO          | ['The Mercado                            |                                                                                                                                                                                                                                                                             | 1           |
| PLAZA PRINCIPAL              | GEO          | ['The Plaza Prin                         |                                                                                                                                                                                                                                                                             | 1           |
| NÚCLEO URBANO                | GEO          | ['The Núcleo U                           |                                                                                                                                                                                                                                                                             | 1           |

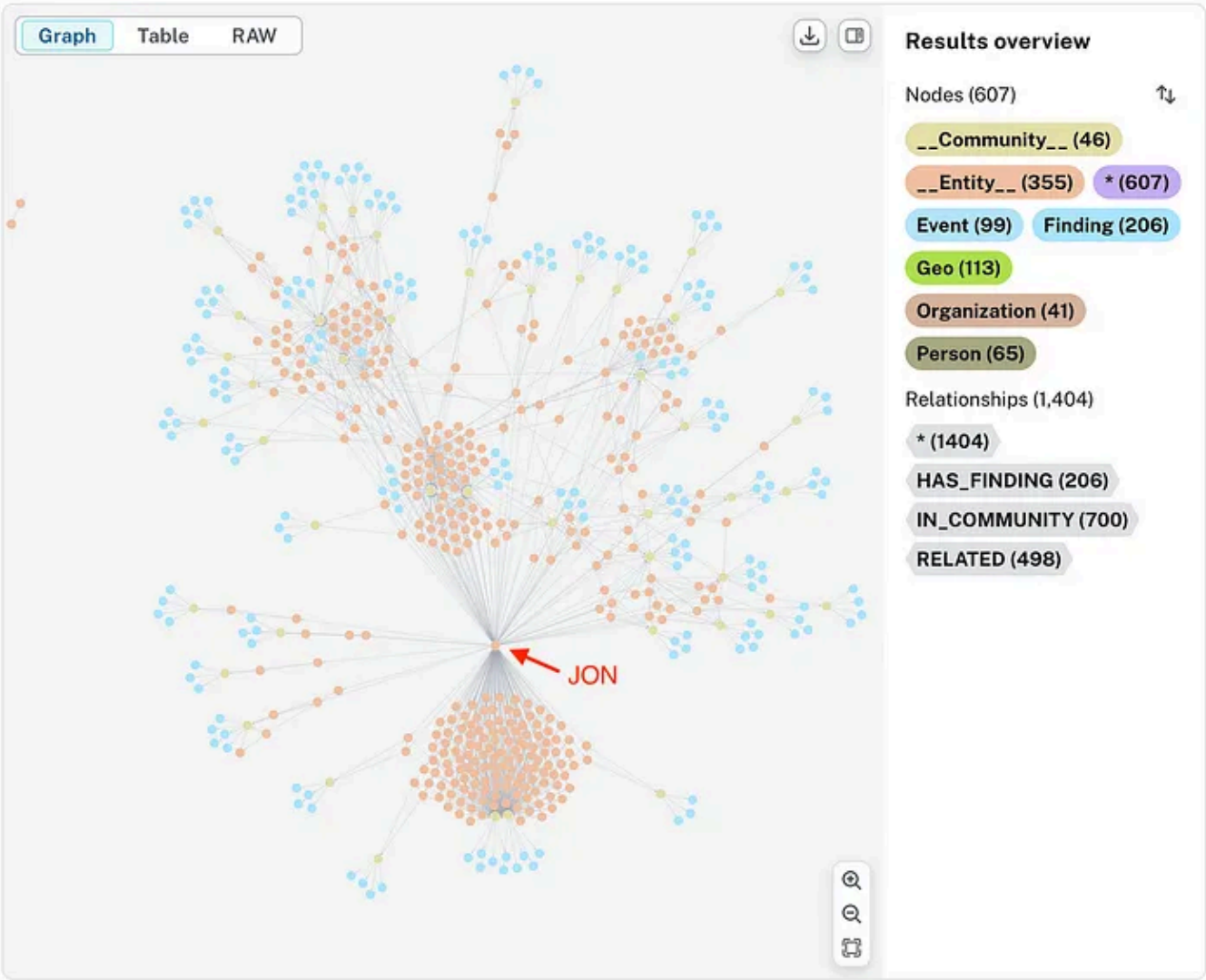
| source     | target           | description                           | text_unit_ids                                                                                                                            | # weight |
|------------|------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------|
| TETUÁN     | MADRID           | ['Tetuán is a neighborhood located v  | ['e13920bd8272086711e7035ca77ea                                                                                                          | 8.0      |
| TETUÁN     | PARRICIDIO       | ['The parricide event occurred in Tet | ['e13920bd8272086711e7035ca77ea                                                                                                          | 1.0      |
| PARRICIDIO | NOTICIAS         | ['The news segment about the parric   | ['e13920bd8272086711e7035ca77ea                                                                                                          | 1.0      |
| JON        | CELIA            | ['Jon feels a mix of emotions regardi | ['9e0705f640b47ed821ec0df09266bf                                                                                                         | 80.0     |
| JON        | MURDER           | ['Jo                                  | ['Jon feels a mix of emotions regarding Celia, the murderer, as he                                                                       | 1.0      |
| JON        | WINDOW           | ['Jo                                  | grapples with the implications of her actions and his own role in the unfolding events', 'Celia is mentioned in Jon's thoughts about the | 13.0     |
| CELIA      | LAURA            | ['Ce                                  | mysterious woman, suggesting a connection between them', 'Jon is concerned that if Laura starts connecting the dots, she might reach     | 60.0     |
| CELIA      | SPAIN            | ['Ce                                  | Celia, indicating a relationship between Jon and Celia', 'Jon's                                                                          | 6.0      |
| CELIA      | MURDER           | ['Th                                  | thoughts about Laura's struggles may parallel his own experiences                                                                        | 9.0      |
| CELIA      | BANK             | ['Ce                                  | with Celia's past, indicating a thematic connection', 'Jon killed Celia                                                                  | 7.0      |
| JON        | TRUMAN CAPOTE    | ['Jo                                  | to protect Laura, believing he was delivering justice', 'Celia appears                                                                   | 5.0      |
| JON        | SATURDAY MORNING | ['Jo                                  | in Jon's nightmare, indicating a significant emotional or psychological                                                                  | 1.0      |
| JON        | JON'S HOME       | ['Jo                                  | connection", "Jon is emotionally affected by Celia's past and her                                                                        | 7.0      |
| JON        |                  |                                       | connection to Laura, which complicates his feelings", 'Jon realizes                                                                      |          |
| JON        |                  |                                       | that Celia is trying to escape, which complicates his feelings about                                                                     |          |

(3) 描述总结与去重：系统会尝试对提取出的信息进行去重。它将同名同类型的实体，以及具有相同源节点和目标节点的关系进行分组。然后，再次调用 LLM，将一个实体或关系在所有文本块中的简短描述汇总，生成一个更全面、更详细的最终描述。

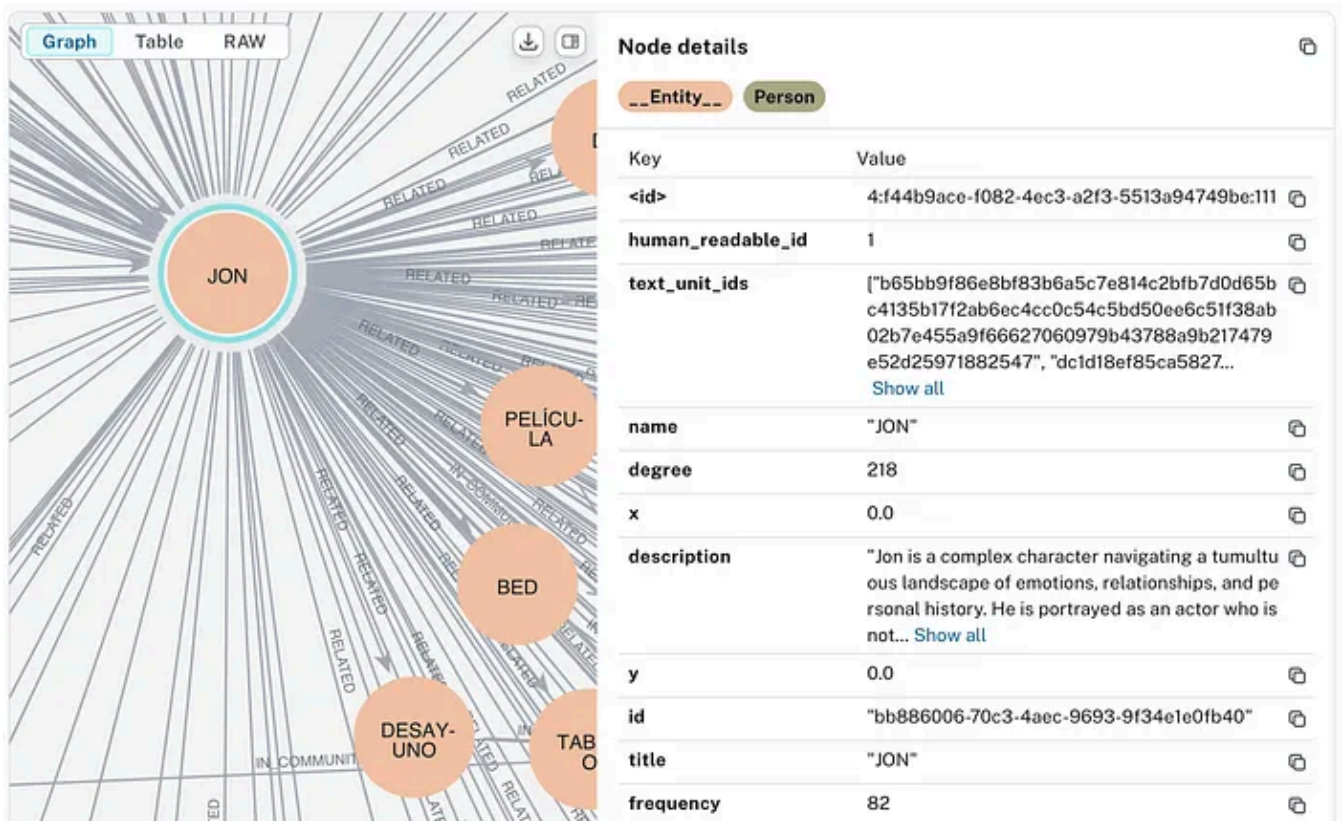
| title                        | type         | text_unit_ids                     | # frequency | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|--------------|-----------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MADRID                       | GEO          | ['b65bb9f86e8bf83b6a5c7e814c2bft] | 6           | Madrid is the capital city of Spain, kr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| JON                          | PERSON       | ['b65bb9f86e8bf83b6a5c7e814c2bft] | 82          | Jon is a complex character navigating a tumultuous landscape of emotions, relationships, and personal history. He is portrayed as an actor who is not only preparing for a role in a miniseries focused on real-life stories of loss and recovery but is also deeply affected by his past experiences, particularly a traumatic event involving a woman named Celia, whose fate haunts him. This incident has left him grappling with feelings of guilt, fear, and paranoia, especially as he deals with unsettling noises from a neighboring house and the implications of his actions. |
| JON                          | PERSON       | ['b65bb9f86e8bf83b6a5c7e814c2bft] | 82          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| LA TRAVIATA                  | EVENT        | ['b65bb9f86e8bf83b6a5c7e814c2bft] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| COCHE DEPORTIVO              | ORGANIZATION | ['b65bb9f86e8bf83b6a5c7e814c2bft] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| REDES SOCIALES               | ORGANIZATION | ['b65bb9f86e8bf83b6a5c7e814c2bft] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| CONTRATO                     | EVENT        | ['b65bb9f86e8bf83b6a5c7e814c2bft] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| AGENCIA DE PUBLICIDAD        |              | ['b65bb9f86e8bf83b6a5c7e814c2bft] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| JULIO                        | PERSON       | ['dc1d18ef85ca582714aa0a57b840c7] |             | Throughout the narrative, Jon experiences significant emotional distress and isolation, reflecting on his life choices and the impact of his work on his identity. He is often anxious about being recognized as a famous actor, which adds to his struggles with privacy and public perception. His relationship with Laura is central to his emotional journey; he shows concern for her well-being and engages in deep conversations about their pasts, revealing a                                                                                                                   |
| JULIO                        | PERSON       | ['dc1d18ef85ca582714aa0a57b840c7] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| CUATRO LATAS                 | GEO          | ['dc1d18ef85ca582714aa0a57b840c7] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| FÉLIX RODRÍGUEZ DE LA FUENTE | PERSON       | ['dc1d18ef85ca582714aa0a57b840c7] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| MERCADO                      | GEO          | ['dc1d18ef85ca582714aa0a57b840c7] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| PLAZA PRINCIPAL              | GEO          | ['dc1d18ef85ca582714aa0a57b840c7] |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

值得注意的是，这种去重机制并不完美，且 GraphRAG 目前不处理实体消歧（例如，“Jon”和“Jon Márquez”会被视为两个独立的节点，即使它们指向同一个人）。

(4) 图的最终形成 ( finalize\_graph )：最后，使用 NetworkX+ 库将所有实体和关系表示为图的节点（Nodes）和边（Edges），并计算节点的度（degree）等结构化信息。  
为了直观理解，下图使用 Neo4j+ 对生成的图谱进行了可视化。







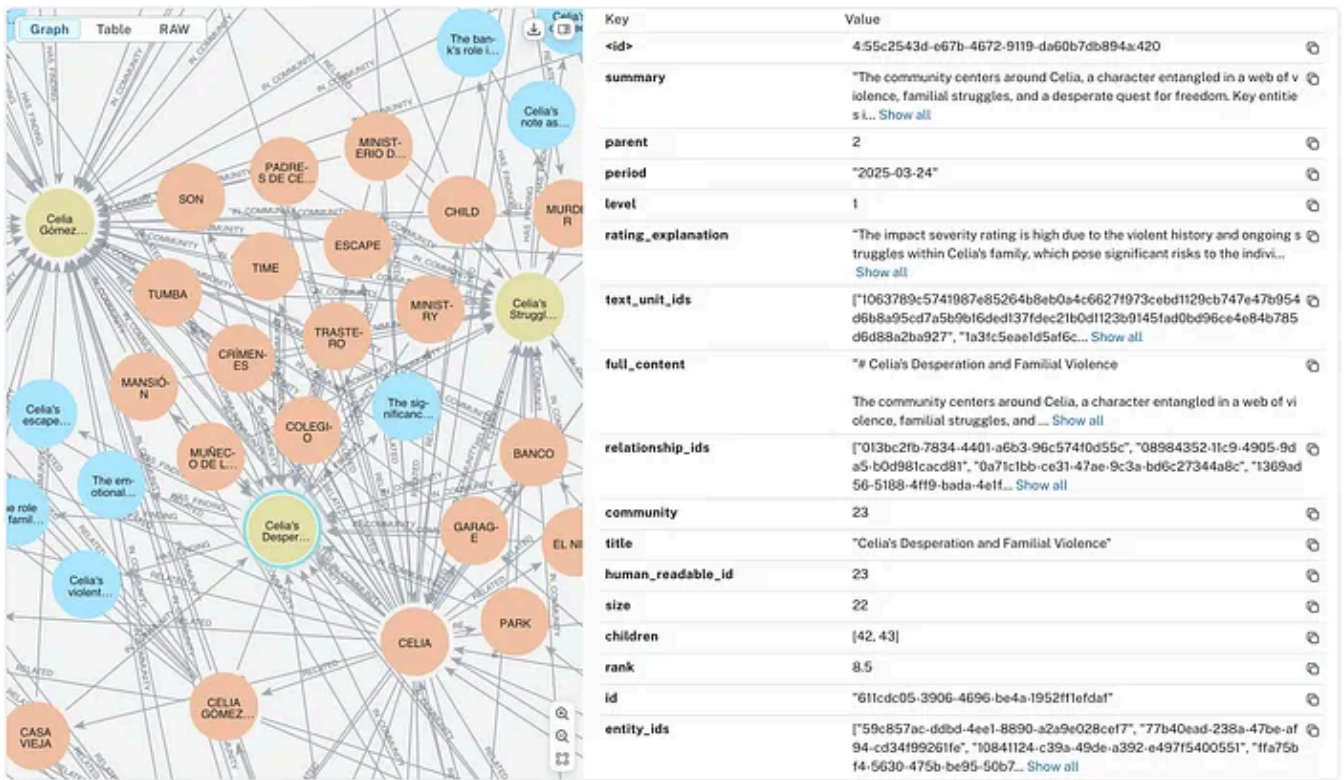
图分区与社区报告

在图构建完成后，GraphRAG会对其进行分区，以发现更高层次的结构化信息。

(1) 社区划分 ( create\_communities )：此步骤使用 [Leiden 算法](#)（一种层级聚类算法）将图划分为多个社区（Communities）。社区是一组内部连接比外部连接更紧密的节点集合。Leiden 算法的层级特性使得社区可以有不同的粒度，级别（level）越高，社区的主题越具体。

| community | level | relationship_ids                              | text_unit_ids | # parent                          | entity_ids | id          | human_readable_id   | title      | children | period | # size |
|-----------|-------|-----------------------------------------------|---------------|-----------------------------------|------------|-------------|---------------------|------------|----------|--------|--------|
| 0         | 0     | ['002dd161-6def-4b65-bf', '04f60d35ff6ea1d1'] | -1            | ['bb886006-70c3-49481fe5-d26']    | 0          | Community 0 | [12 13]             | 2025-03-28 |          | 131    |        |
| 1         | 0     | ['0179caa7-01ac-4af2-95', '0432d276c182aa2f'] | -1            | ['fb50f1da-c96e-45cfb4792-a54']   | 1          | Community 1 | [14 15 16]          | 2025-03-28 |          | 66     |        |
| 2         | 0     | ['0301ddde-c146-4fb0-94', '1063789c5741987f'] | -1            | ['48395483-164e-4c2d3d8f6-265']   | 2          | Community 2 | [17 18 19 20 21 22] | 2025-03-28 |          | 45     |        |
| 3         | 0     | ['00b1c81b-2848-4837-a', '6e0f12be26e0548f']  | -1            | ['458a77bb-2b4f-4575a249b-edf']   | 3          | Community 3 | [23 24 25 26 27]    | 2025-03-28 |          | 21     |        |
| 4         | 0     | ['14aab325-1dbb-4599-8', '09961e5117378bc']   | -1            | ['80937f7d-f094-4cd294dfdf-84bf'] | 4          | Community 4 | [28 29]             | 2025-03-28 |          | 10     |        |

社区的价值在于它们能够整合来自不同实体和关系的信息，从而揭示宏观主题。在书籍分析中，社区可以对应书中的主要情节或主题。



(2) 生成社区报告 ( create\_community\_reports )：系统会为每个社区调用 LLM 生成一份详细的报告，概述其主要事件或主题。LLM 的上下文包含了该社区内的所有实体、关系和声明 (claims)。

对于大型社区，上下文可能超出 LLM 输入长度限制。此时，GraphRAG 采用两种策略缩减上下文：

- 层级替换 (Hierarchical Substitution)：用子社区的报告来替代其内部的实体和关系等原始文本，优先替换包含实体最多的子社区。
- 裁剪 (Trimming)：如果替换后上下文仍然过长，则根据节点度等指标移除最不重要的实体和关系。

最终，LLM 会生成一份包含关键洞见 (findings) 和摘要 (summary) 的社区报告。

| summary                                                                                                                                                                                                                                                                                                                                                                                                             | full_content                                   | #  | rank | rating_explanation                                                                                                                                                      | findings                                                                                                                                                                                 | full_content_json                                                                                                              | period     | #   | size |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------|-----|------|
| The community centers around Jon, a character depicted as an actor who is deeply affected by his public image, which is shaped by his contract with an advertising agency and his social media presence.                                                                                                                                                                                                            | # Jon's Emotional Landscape and Public Persona | 1  | 7.5  | The impact severity rating is high due to the violent history and ongoing struggles within Celia's family, which pose significant risks to the individual's well-being. | [[{"explanation": "Jon is depicted as an actor who is deeply affected by his public image, which is shaped by his contract with an advertising agency and his social media presence."}]] | ["title": "Jon's Emotional Landscape and Public Persona", "content": "# Jon's Emotional Landscape and Public Persona"]         | 2025-03-28 | 129 |      |
| The community is centered around the village, a complex character navigating his public image as an actor while grappling with personal trauma and emotional turmoil. Key entities include his relationships, his home environment, and various cultural references that shape his identity. The interconnectedness of these elements highlights the challenges Jon faces in reconciling his past with his present. | # Jon's Emotional Landscape and Public Persona | 2  |      | severity rating is moderate                                                                                                                                             | [[{"explanation": "The Village serves as a significant cultural reference point for Jon, highlighting his complex relationship with his home environment."}]]                            | ["title": "Village and Night: A Tension", "content": "# Jon's Emotional Landscape and Public Persona"]                         | 2025-03-28 | 2   |      |
| The community centers around the garage, a complex character navigating his public image as an actor while grappling with personal trauma and emotional turmoil. Key entities include his relationships, his home environment, and various cultural references that shape his identity. The interconnectedness of these elements highlights the challenges Jon faces in reconciling his past with his present.      | # Jon's Emotional Landscape and Public Persona | 62 |      | severity rating is high due                                                                                                                                             | [[{"explanation": "Jon's experiences are shaped by his complex relationship with his home environment, which is a significant cultural reference point for him."}]]                      | ["title": "Jon and Laura's Complex Relationship", "content": "# Jon's Emotional Landscape and Public Persona"]                 | 2025-03-28 | 62  |      |
| This community is characterized by a significant cultural reference point for Jon, highlighting his complex relationship with his home environment, which is a significant cultural reference point for him.                                                                                                                                                                                                        | # Jon's Emotional Landscape and Public Persona | 2  |      | severity rating is low due                                                                                                                                              | [[{"explanation": "The garage is a significant cultural reference point for Jon, highlighting his complex relationship with his home environment."}]]                                    | ["title": "Garage and Camara Frigorífica", "content": "# Jon's Emotional Landscape and Public Persona"]                        | 2025-03-28 | 2   |      |
| The community centers around Celia, a complex character navigating his public image as an actor while grappling with personal trauma and emotional turmoil. Key entities include his relationships, his home environment, and various cultural references that shape his identity. The interconnectedness of these elements highlights the challenges Jon faces in reconciling his past with his present.           | # Jon's Emotional Landscape and Public Persona | 2  |      | severity rating is high due                                                                                                                                             | [[{"explanation": "The entity NOTA is a significant cultural reference point for Jon, highlighting his complex relationship with his home environment."}]]                               | ["title": "Community of Urgency: NO", "content": "# Jon's Emotional Landscape and Public Persona"]                             | 2025-03-28 | 2   |      |
| This community centers around the garage, a complex character navigating his public image as an actor while grappling with personal trauma and emotional turmoil. Key entities include his relationships, his home environment, and various cultural references that shape his identity. The interconnectedness of these elements highlights the challenges Jon faces in reconciling his past with his present.     | # Jon's Emotional Landscape and Public Persona | 25 |      | severity rating is high due                                                                                                                                             | [[{"explanation": "Celia Gómez Saavedra and her family are a significant cultural reference point for Jon, highlighting his complex relationship with his home environment."}]]          | ["title": "Celia Gómez Saavedra and her Family", "content": "# Jon's Emotional Landscape and Public Persona"]                  | 2025-03-28 | 25  |      |
| This community centers around the garage, a complex character navigating his public image as an actor while grappling with personal trauma and emotional turmoil. Key entities include his relationships, his home environment, and various cultural references that shape his identity. The interconnectedness of these elements highlights the challenges Jon faces in reconciling his past with his present.     | # Jon's Emotional Landscape and Public Persona | 10 |      | severity rating is high due                                                                                                                                             | [[{"explanation": "José's struggle with his public image is a significant cultural reference point for Jon, highlighting his complex relationship with his home environment."}]]         | ["title": "José, Celia, and the Struggle with Public Perception", "content": "# Jon's Emotional Landscape and Public Persona"] | 2025-03-28 | 10  |      |
| This community centers around the garage, a complex character navigating his public image as an actor while grappling with personal trauma and emotional turmoil. Key entities include his relationships, his home environment, and various cultural references that shape his identity. The interconnectedness of these elements highlights the challenges Jon faces in reconciling his past with his present.     | # Jon's Emotional Landscape and Public Persona | 2  |      | severity rating is high due                                                                                                                                             | [[{"explanation": "Murder is a pivotal event in the community's history, highlighting the challenges Jon faces in reconciling his past with his present."}]]                             | ["title": "Murder Investigation and the Community's Response", "content": "# Jon's Emotional Landscape and Public Persona"]    | 2025-03-28 | 2   |      |
| The community centers around the television, a complex character navigating his public image as an actor while grappling with personal trauma and emotional turmoil. Key entities include his relationships, his home environment, and various cultural references that shape his identity. The interconnectedness of these elements highlights the challenges Jon faces in reconciling his past with his present.  | # Jon's Emotional Landscape and Public Persona | 2  |      | severity rating is elevated                                                                                                                                             | [[{"explanation": "The television serves as a significant cultural reference point for Jon, highlighting his complex relationship with his home environment."}]]                         | ["title": "Television and Ministry Influence", "content": "# Jon's Emotional Landscape and Public Persona"]                    | 2025-03-28 | 2   |      |

(3) 生成嵌入向量 ( generate\_embeddings )：最后一步，系统会为所有文本单元、实体描述和社区报告（包含标题、摘要、报告全文和评级）创建嵌入向量 (Embeddings)。这些向量将在后续的查询阶段用于高效的语义搜索。

## 第二阶段：图查询

图构建完成后，便可以对其进行查询。GraphRAG 提供了两种主要的查询方法：局部搜索和全局搜索。

### 局部搜索 (Local Search)：应对具体问题

当你需要回答一个具体、明确的问题时，应使用局部搜索。

查询示例： "Laura 在寻求什么样的报复，为什么？ "

## 局部搜索 (Local Search)



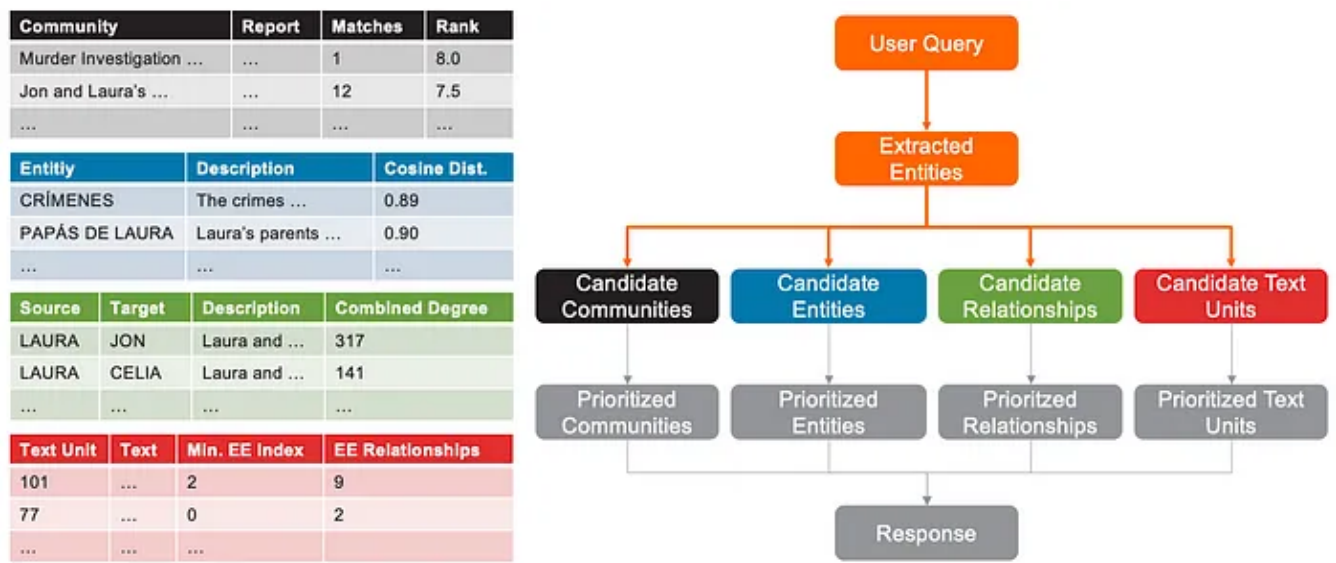
局部搜索的流程如下：

(1) 检索相关实体：首先，系统将用户查询嵌入为向量，并与知识库中所有实体的描述向量进行语义相似度计算，找出最相似的N个实体（由 top\_k\_mapped\_entities 参数控制）。

| id           | short_id | title          | type   | description                           | description_embed... | name_em...    | community_ids | text_unit_ids           | # | rank | attributes    |
|--------------|----------|----------------|--------|---------------------------------------|----------------------|---------------|---------------|-------------------------|---|------|---------------|
| 869ae4a8-a6f | 317      | CRIMENES       | EVENT  | The crimes committed by Laura are     | Missing value        | Missing value | ['1', '14']   | ['cb6ff6db7b62e9c78c99  |   | 2    | Missing value |
| bb179169-3e  | 318      | PAPÁS DE LAURA | PERSON | Laura's parents are referenced as hav | Missing value        | Missing value | ['1', '14']   | ['cb6ff6db7b62e9c78c99  |   | 1    | Missing value |
| 6d0fee65-6a7 | 130      | LAURA          | PERSON | Laura is a complex character deeply   | Missing value        | Missing value | ['1', '14']   | ['4951758edd584740d50   |   | 99   | Missing value |
| e8037f05-f4d | 367      | KNIFE          | EVENT  | The knife is the weapon used by Lau   | Missing value        | Missing value | ['1', '14']   | ['f0aa4da8a6af89ece236f |   | 1    | Missing value |
| 9b5d7626-15f | 218      | MURDER         | EVENT  | The act of murder is a pivotal eleme  | Missing value        | Missing value | ['2', '19']   | ['9e0705f640b47ed821ec  |   | 5    | Missing value |

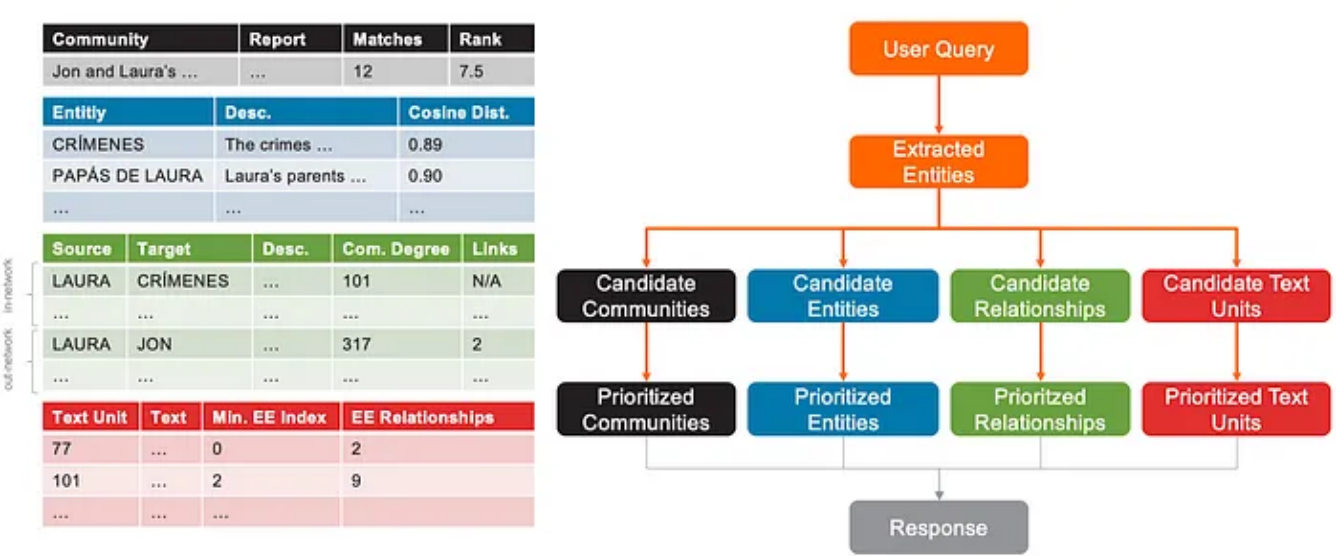
(2) 构建候选集：这N个实体被称为候选实体。系统会围绕这些实体进行扩展，构建出四个候选集：

- 候选社区：所有包含至少一个候选实体的社区。
- 候选关系：所有以候选实体为源节点或目标节点的边。
- 候选文本单元：所有包含至少一个候选实体的原始文本块。



(3) 上下文排序与筛选：由于LLM的上下文窗口有限，必须对候选集进行排序，以确保最重要的信息被优先送入模型。排序规则非常精细：

- 社区排序：按社区内候选实体出现的文本单元数量（matches）降序排列，若相同则按社区重要性（rank）排序。
- 关系排序：网络内关系（连接两个候选实体）优先于网络外关系（连接一个候选实体和非候选实体）。网络内关系按组合度（combined\_degree，源和目标节点度之和）排序；网络外关系则按其连接到网络内实体的数量排序。
- 文本单元排序：首先按照其关联的候选实体的语义相似度顺序排序，其次按关联的关系数量排序。



(4) 生成最终答案：系统 将经过排序和筛选后的社区信息、实体描述、关系描述和文本单元拼接成一个完整的上下文，并将其与用户问题一起提交给LLM

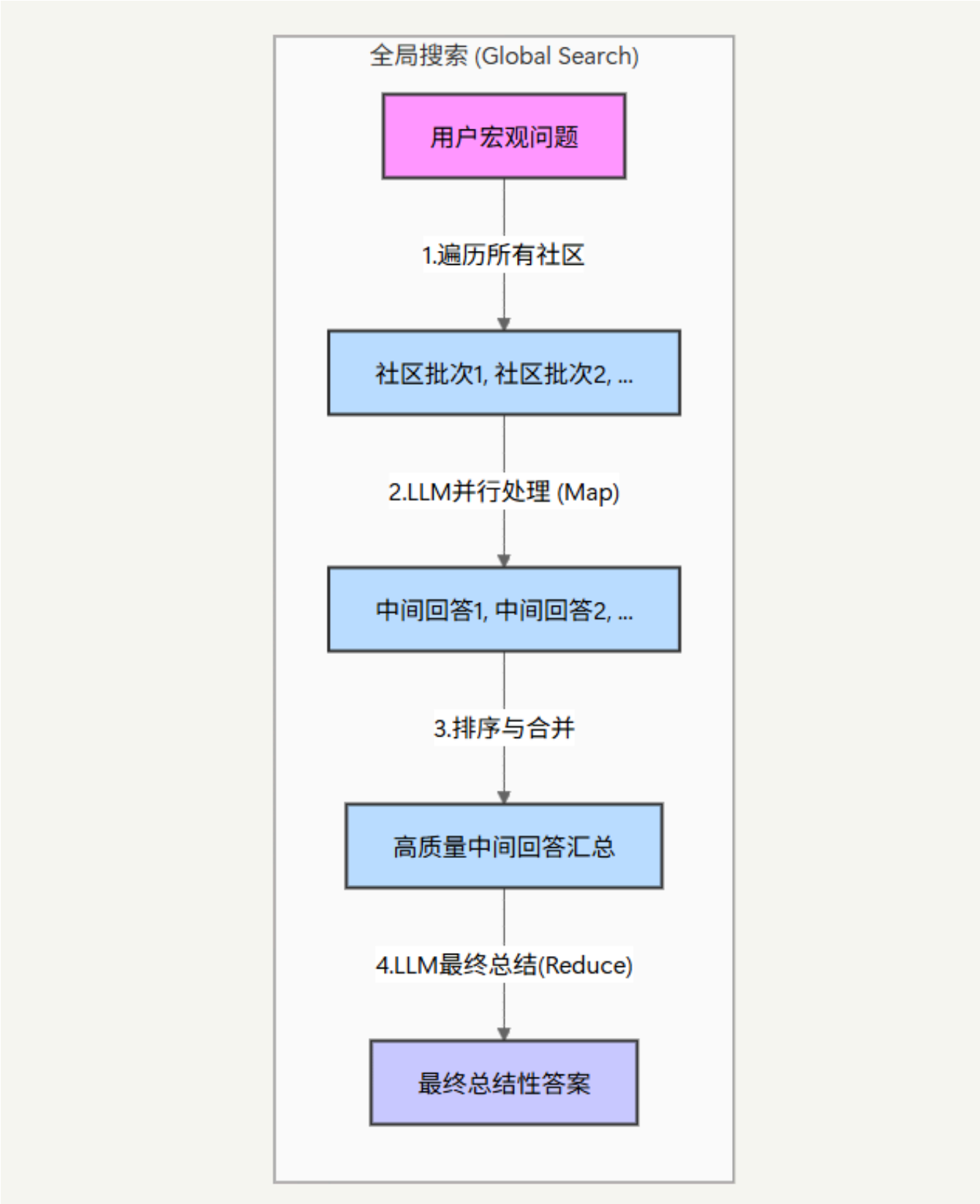


全局搜索 (Global Search)：探索宏观主题

当你想了解宽泛、总结性的主题时，应使用全局搜索。

查询示例： "这本书探讨了哪些主题？"

全局搜索采用了一种类似 MapReduce+ 的策略来处理问题。



全局搜索的流程如下：

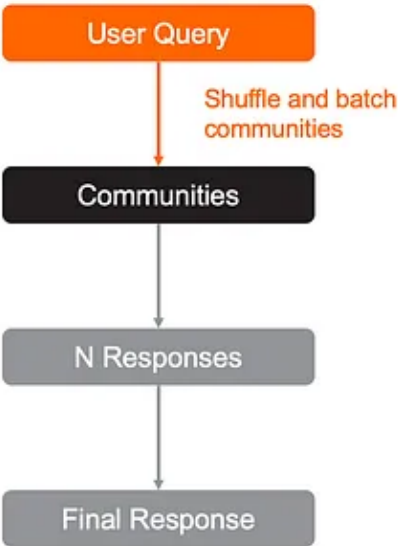
(1) 社区加权与分片：首先，社区权重与分片大小由社区质量得分决定。社区质量得分由社区内高质量回答的数量、社区内高质量回答的多样性、社区内高质量回答的互动性等因素决定。然后，所有社

"What themes are explored in the book?"

| ID  | Community                           | Report | Comm. Weight |
|-----|-------------------------------------|--------|--------------|
| 2   | Celia, Laura, and the Tragic Murder | ...    | 1.0          |
| 21  | Trastero and Sótano Community       | ...    | 0.01         |
| ... | ...                                 | ...    | ...          |

| ID  | Community                     | Report | Comm. Weight |
|-----|-------------------------------|--------|--------------|
| 1   | Jon's Emotional Landscape ... | ...    | 1.00         |
| ... | ...                           | ...    | ...          |

| ID  | Community                           | Report | Comm. Weight |
|-----|-------------------------------------|--------|--------------|
| 6   | Jon Márquez and the Dynamics of ... | ...    | 0.39         |
| ... | ...                                 | ...    | ...          |



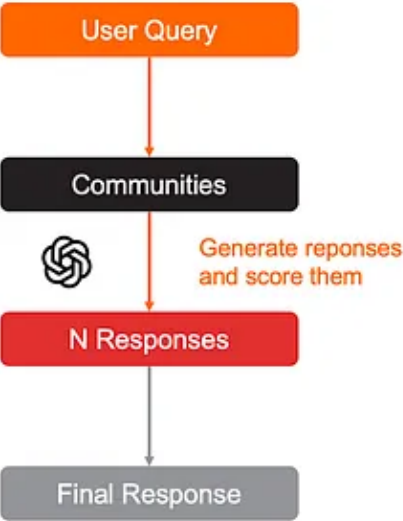
(2) 生成中间回答 (Map阶段): 对于每个社区批次, LLM会以该批次的社区报告为上下文, 生成多个对用户问题的回答, 并为每个回答打分 (0-10分), 以评估其相关性。

"What themes are explored in the book?"

Batch 0

| ID  | Community                           | Report | Comm. Weight |
|-----|-------------------------------------|--------|--------------|
| 2   | Celia, Laura, and the Tragic Murder | ...    | 1.0          |
| ... | ...                                 | ...    | ...          |
| 21  | Trastero and Sótano Community       | ...    | 0.01         |

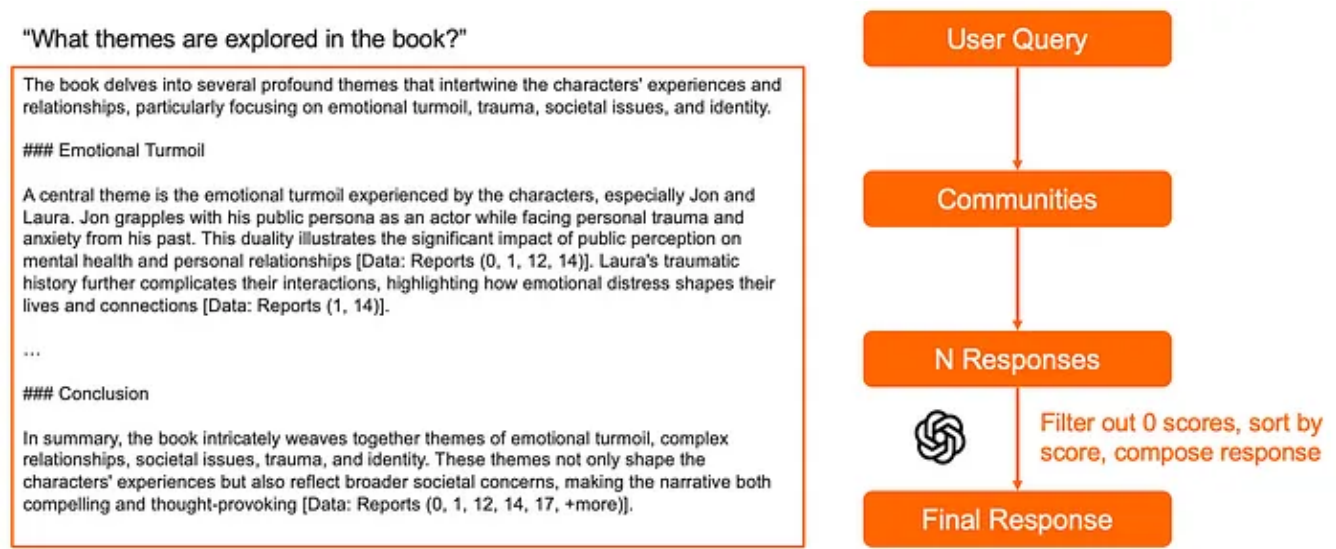
| Batch | Response                                               | Score |
|-------|--------------------------------------------------------|-------|
| 0     | The theme of emotional turmoil is prevalent ...        | 90    |
| 0     | The exploration of identity and self-perception is ... | 70    |
| ...   | ...                                                    | ...   |



(3) 排序与汇总: 所有生成的中间回答会根据分数进行全局排序, 分数过低的回答被舍弃。

| # | analyst | answer                                                           | # | score |
|---|---------|------------------------------------------------------------------|---|-------|
|   | 0       | The theme of emotional turmoil is prevalent, particularly in tl  |   | 90    |
|   | 1       | The theme of emotional turmoil is prevalent, particularly in tl  |   | 90    |
|   | 2       | The theme of emotional turmoil is explored through Jon Már       |   | 90    |
|   | 0       | The narrative explores the significance of relationships, partic |   | 85    |
|   | 1       | The narrative explores complex relationships, particularly thro  |   | 85    |
|   | 1       | The theme of trauma is explored through the shared experie       |   | 85    |
|   | 2       | The complexities of relationships are central to the narrative,  |   | 85    |
|   | 0       | Societal issues, such as domestic violence and the impact of     |   | 80    |
|   | 1       | Societal issues such as gender-based violence and crime are      |   | 80    |
|   | 2       | The significance of place is examined through the Village and    |   | 80    |
|   | 0       | The theme of trauma and its lasting effects is explored throu    |   | 75    |
|   | 1       | The influence of media and public perception is a recurring t    |   | 75    |
|   | 2       | The influence of media on personal identity is a key theme, a    |   | 75    |
|   | 0       | The exploration of identity and self-perception is significant,  |   | 70    |
|   | 2       | The interplay between night and day serves as a thematic ele     |   | 70    |

(4) 生成最终答案 (Reduce阶段): 最后, 系统将所有高质量的中间回答文本拼接成一个新的上下文, 再次提交给 LLM, 要求其基于这些信息生成一个全面、连贯的最终答案。



总结

本文详细拆解了微软 GraphRAG 从图构建到两种核心查询模式（局部搜索与全局搜索）的完整工作流程。通过将非结构化文本转化为结构化的知识图谱，并利用社区划分、层级化报告和精细的上下文管理策略，GraphRAG 能够有效地回答从具体细节到宏观主题的各类问题，为 RAG 技术的发展提供了强大的新思路。希望本文能够帮助你深入理解其背后的机制，并启发你在自己的项目中应用 GraphRAG 的强大能力。

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